

AVINA

New York, NY

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Education

New York University

MS Computer Science, GPA: 3.6

May 2025

New York, USA

Vellore Institute of Technology

B.Tech Computer Science and Engineering, GPA: 8.89

June 2021

Vellore, India

Experience

Metropolitan Transportation Authority (MTA)

Feb 2025 – May 2025

Operations Technology Intern

New York

- Develop a Workforce Management System using Python leveraging historical data and forecasting techniques to predict demand, optimize workload allocation, and minimize overtime by identifying inefficiencies

Develop for Good

October 2024 – Current

Engineering Manager Intern

New York

- Led the cross platform mobile development using React Native, Typescript, Supabase and Expo
- Define technical milestones and weekly deliverables to ensure a timely delivery and seamless experience of the app

PharmEasy

July 2021 – July 2023

Software Developer

Bangalore

- Managed microservices architecture on Kubernetes, Spring Boot with Java and Kotlin, ensuring system reliability
- Architected scalable distributed systems to optimize warehouse operations, reducing TAT by 37% and operational costs by 20%
- Resolved critical on-call issues, reducing downtime by 30% and e-waybill failure rates from 35% down to 3%
- Integrated Kafka for efficient data streaming, real-time data processing and reducing message processing latency
- Optimized high-throughput SQL databases, reducing query response time to <10 seconds minimizing service disruptions

ISRO-Space Applications Centre

Dec 2020 – May 2021

Research Intern - Image Processing

Remote

- Designed and executed a novel system for Incidence Angle Normalization of Ocean ENVISAT satellite images using Python, thereby reducing amplitude by a factor of 5 and decreasing backscatter by 10dB
- Executed data pre-processing and Image Processing on large Radar and Geospatial data like N1, Geotiff, GIS, ENVI
- Built Java GUI and integrated four other Satellite images processing projects, enhancing user experience

Amilcar Technologies

Sep 2020 – Nov 2020

Web Development Intern

Remote

- Led redesign and development of the official company website, and organizing company information
- Used HTML, CSS, JavaScript, PHP, Bootstrap, and jQuery, for website development ensuring cross-browser and cross-device compatibility for a seamless user experience and accessibility

Projects

Classmatelo | AWS, Python, Natural Language Processing, Flask, ML

December 2023

- Scalable web application designed on AWS, to revolutionize the learning experience using gamification
- Integrated S3, Lambda, CodeFormation, DynamoDB, Lex, Transcribe and deployed Flask app on EC2 for high traffic handling and auto scaling, to implement lecture audio transcription, quiz-based learning, calendar events and reminders
- Utilized NLP techniques for text summarization and MCQ generation

StepFreeMTA | Django, AWS, Google Maps API, Javascript

December 2024

- Developed a web-based subway app to assist individuals with disabilities and mobility challenges
- Using Django, deployed on AWS Elastic Beanstalk, leveraging Google Maps API, including Places, Search, and Routing APIs
- Integrated trip planning, real-time updates from MTA and NYCOpenData, and crowdsourced reporting for reliability
- Ensured code quality and coverage with tools like Coveralls

Dining Concierge | AWS, Python, Web scraping, NLP

October 2023

- Serverless, microservice driven chatbot, that sends restaurant suggestions given a set of preferences
- Built on AWS; the data for restaurants are scraped from yelp from 5,000+ random restaurants from Manhattan

Skills

Java, Kotlin, Python, C, C++, Linux, Shell, Virtualization, Data Structures and Algorithms, GIS, Data Analytics

Website and App : SpringBoot, RestApi, AWS, HTML, CSS, JavaScript, jQuery, PHP, React, Bootstrap, SQL, Git, Kubernetes, Docker, Kafka, MySQL, PostgreSQL, MongoDB, NodeJS, Android Studio, Agile, TypeScript, Django, Flask

Artificial Intelligence:: NLP, GenAI, PyTorch, Scikit, Pandas, Machine learning, Tensorflow, Deep Learning